Actual all-node moments and whole-chart precision

14 November 2026; 38 authored Lean statements
Source `3299b929faad599061b1c13eed9de02ef7e806f2`

Construction. For the same completed auxiliary orientation polynomial, the new component constructs first-jet moment factors at every retained nonzero source node, then derives a finite geometric approximation for every coefficient of its actual crossing chart. The approximation is per entry; compact whole-polynomial matrix factors are a separate open obligation. All records are unsigned [pcc:unverified].

Actual inputs. Let $L = 2^a$, $S = \{-L, \dots, L-1\} \subset \mathbb{Q}$, $t \in S \cap \mathbb{Z}$ nonzero with $v_2(t) = i < u \le a$, and actual ports $x, y < L$ with equal bit i. Use natural multiplicities m_j, the same completed $W_{xy}(T)$, and

$$r = x \bmod 2^u, \quad h = x \operatorname{div} 2^u, \quad \delta_x = y - x, \quad \delta_r = y - r, \quad \alpha = 2^{a+1}.$$

The reference c is the existing nonzero diagonal first jet at the negative pole -2^i, with valuation $w = a + 1 - i + G_i$, where $G_i = m_i - \sum_{j<u}(j-i)_+ m_j$. It is not the potentially zero local first jet at t. Define $P_x(Z) = W_{xy}(t + \alpha Z)/c$ and P_r similarly. The previous all-node source derives $P_x(0) = 0$ and normalized coefficient $n+1$ of weight at least n, hence integral whole polynomials [pcc:unverified].

First-jet moments. Literal prototype denominators give even slopes

$$e_t = \frac{2^u}{t - \delta_r}, \qquad e_j = \frac{2^u}{2^j - \delta_r}, \qquad D_H(H) = (1 + e_t H) \prod_{j \in M_{xy}} (1 + e_j H)^{m_j}.$$

Its constant is one and coefficient b has valuation at least b. The actual normalized first-jet family is its prototype first jet times $D_H(h)^{-1}$. Standard inverse-series truncation below k yields constructed $2^u k$ factors, left weight b, integral right, and normalized residual k. Actual c scaling gives left weight $w + b$ and residual $w + k$. No positive-multiplicity or nonzero local-jet assumption enters this profile [pcc:unverified].

At a positive retained anchor with $m_i > 0$, every actual first crossing jet and corresponding right moment factor is zero. This control does not make the higher polynomial chart zero [pcc:unverified].

Actual polynomial denominator and geometric error

Derive the identity from W. Exact retained-mask comparison and the original nodal puncture identity give

$$(T-\delta_x)W_{xy}(T) = q(T-\delta_r)W_{ry}(T), \qquad q = \prod_{j\in M_{xy}} \left(\frac{2^j-\delta_r}{2^j-\delta_x}\right)^{m_j}.$$

Only actual nonzero denominators are inverted. No local first jet is cancelled. Put $d_x = t-\delta_x$, $d_r = t-\delta_r$ and $\lambda = \alpha/d_r$. Construct

$$D(Z) = q^{-1}\left(\frac{d_x}{d_r} + \lambda Z\right), \qquad N(Z) = (1+\lambda Z)P_r(Z).$$

The literal row identity proves $DP_x = N$. Actual crossing-bit denominators give $q^{-1} \equiv d_x/d_r \equiv 1 \pmod 2$ and $v_2(\lambda) \geq 1$. Thus $D-1$ has content at least one and P_x, N are integral. These facts are proved for the actual inputs, not supplied as favorable premises [pcc:unverified].

Finite construction. Define

$$S_k(D) = \sum_{n=0}^{k-1}(1-D)^n, \qquad A_{x,k} = NS_k(D).$$

The finite geometric identity is $DS_k = 1-(1-D)^k$. If $D-1$ has content one, multiplication by D preserves a desired finite content bound on an integral polynomial: starting at content zero, repeatedly write $R = DR-(D-1)R$ to increase the bound to k. This yields

$$\mathrm{Content}_k(P_x - A_{x,k}), \qquad \mathrm{Content}_{w+k}\big(W_{xy}(t+\alpha Z) - cA_{x,k}(Z)\big).$$

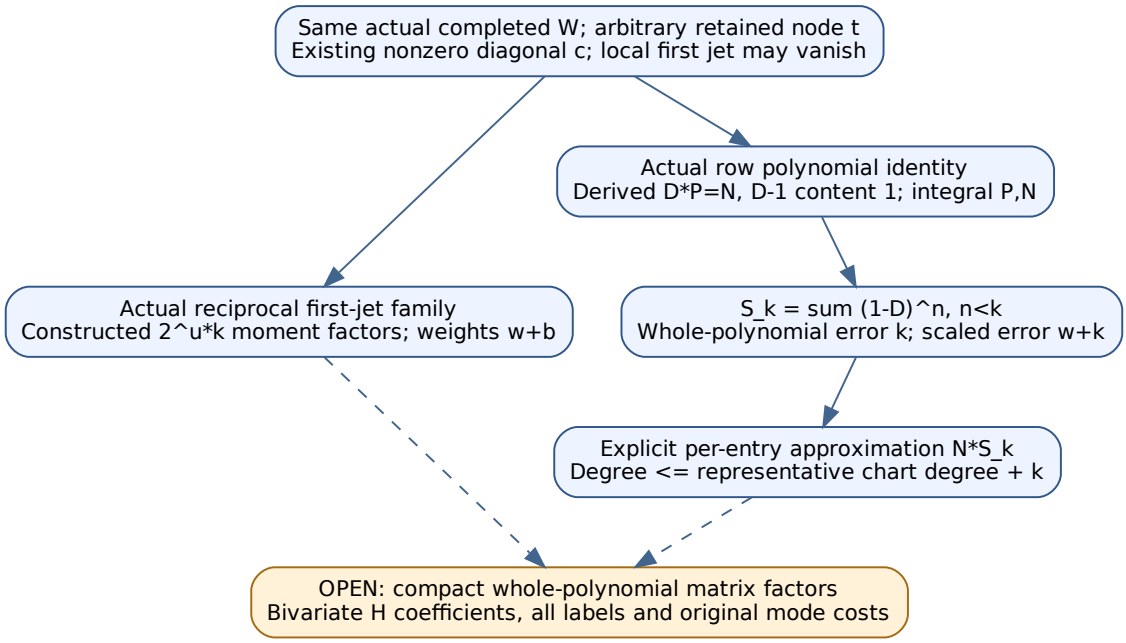
Here content means the bound holds for every polynomial coefficient, including all higher chart coefficients. It permits a vanishing local first jet and uses no inverse of it [pcc:unverified].

Degree accounting. The actual D is affine in Z. The proof bounds $\deg S_k \leq k-1$ (using natural degree at $k=0$) and

$$\mathrm{natdeg}\, A_{x,k} \leq \mathrm{natdeg}\, P_r + k.$$

The $k=0$ approximation is zero and is handled separately. This bounds polynomial degree, not coefficient bit sizes, construction time or storage [pcc:unverified].

Two constructions and the compact-factor gap



The $2^u k$ matrix factors concern the *first jet*. The full-polynomial approximation $A_{x,k}$ still depends on the individual row through q and d_x. No compact full-polynomial matrix factorization is claimed by this component [pcc:unverified].

Next discriminating construction. Write the actual row quotient as H, construct the polynomial denominator

$$\mathcal{D}(H, Z) = (1 + e_t H + \lambda Z) \prod_{j \in M_{ry}} (1 + e_j H)^{m_j},$$

and its finite geometric approximation. Prove coefficient-H^b content at least b, then truncate $b \geq k$ with an actual error bound at integer h. This is a prospective route to compact whole-polynomial factors; its coefficient bridge and matrix product are not part of these 38 statements.

Every residue/moment label must be charged. The original primary interface allows at most $6M/D_i$ modes per weight, $D_i = L/2^i$; a 2^u inventory can exceed its old 2^{i+1} tile count. Full coefficient construction, entry/bit/degree/precision/recove and public relation/finder/ field-scalar extraction/retry/expected cost remain explicit obligations.

Source bindings, validation and preserved failures

The new isolated project builds all 38 authored statements without warnings. Pinned shared gate and import-inclusive fresh replay are pending at this report draft; a running job is not a pass. Six central printed dependencies contain only `propext`, `Classical.choice`, `Quot.sound`. All source records are unsigned [pcc:unverified].

`actual_node_moment_profile`	PCC-20261114-ae019d
`actual_row_denominator_identity`	PCC-20261114-b2de93
`actual_whole_chart_precision`	PCC-20261114-1802a0
`actual_weighted_whole_chart_precision`	PCC-20261114-96a569
`actual_approximation_degree`	PCC-20261114-7649ed

Full statements govern their hypotheses: see `../evidence/claim-pins.json`, source/configuration pins in `../evidence/source-freeze.json`, and prospective contract `../PROTOCOL.md`. Local draft007 data is PCC-20261114-496991; that observation alone is not independent gate/replay coverage [pcc:unverified].

Reproducibility. Lean4.35.0; CryptoLean `148f5dc389b341401e06d76abe7783c4ccf42f2e`; Mathlib `cfc55fc58714a59dc06e6e`
The source is an exact copy of accepted draft007. All earlier failed sources, eight build logs/exits and local data receipts remain under `../../nodes/moment-evidence/`. Failures include private-import visibility, wrong Lake working directory, an unreduced class port, a deprecated API, and two exact argument/wrapper mismatches. No linter was disabled or desired content/factor witness added.

Sources checked. KB unavailable as journaled. Read exact parent Node/Moment/Content/Anchor denominator, inverse-series, graded entry and row-polynomial proofs; pinned Mathlib geometric-sum and polynomial-degree APIs; earlier primary D.4.3–7 and contraction precision/cost reviews. Standard finite geometric sums and inverse-series moments are reused without a novelty, priority, fastest-method or measured speed assertion.

No native crypto, attack, F4/msolve, timing, scaling, paid or cloud experiment ran. The original norm fixture remains unlaunched under its unchanged checkpoint transport blocker. No measured result updates canonical IC graphs/leaderboards; the diagram records proof dependencies.

The parent arbitrary-node42/44/fresh0 project and all90 artifact pins remain exact, as do earlier parent checkpoints. The protected concurrent PCC histories remain unresolved; auto-merge stays off. No claim rewrite, rechain, side selection, signature, prefix waiver or force push is attempted.